

1.

List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

1. Motherboard

- **Function:** The primary printed circuit board that connects all hardware components (CPU, RAM, GPU, Storage) and enables them to communicate with each other.

2. CPU (Central Processing Unit)

- **Function:** Known as the "brain" of the computer, it executes instructions, performs mathematical calculations, and manages the overall system tasks.

3. RAM (Random Access Memory)

- **Function:** High-speed temporary memory that stores active data and open applications for quick access by the CPU while the system is turned on.

4. Storage (SSD / HDD)

- **Function:** The permanent memory drive where the operating system, installed software, games, and personal files are stored long-term.

5. GPU (Graphics Processing Unit / Expansion Card)

- **Function:** A dedicated processing unit designed to handle visual rendering, outputting graphics to the display monitor for games and video content.

6. Power Supply Unit (PSU)

- **Function:** Converts high-voltage AC electricity from the wall outlet into low-voltage DC power required by all internal PC components.

7. Cooling System (CPU Cooler / Case Fans)

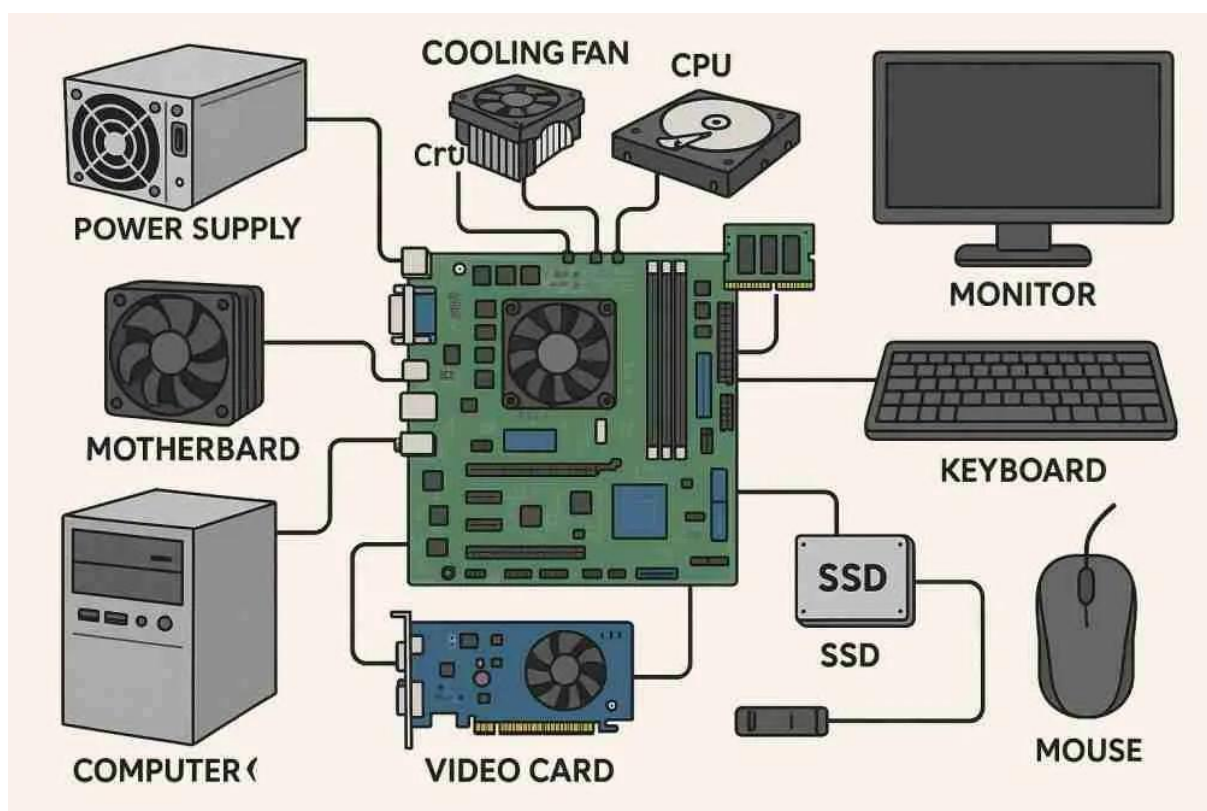
- **Function:** Dissipates heat generated by high-performance components like the CPU and GPU to keep system temperatures within safe operational limits.

2.

Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

Question 1: Main PC Parts and What They Do



☐ **Motherboard**

- **Function:** The primary circuit board that connects all PC parts together so they can share information and communicate.

☐ **CPU (Processor)**

- **Function:** The "brain" of the computer that handles commands, calculations, and runs programs.

☐ **RAM (Memory)**

- **Function:** Fast short-term memory that holds open applications for quick access while the PC is on.
- **Storage (SSD / Hard Drive)**
 - **Function:** Long-term memory where Windows, programs, games, and personal files are permanently saved.
- **GPU (Graphics Card)**
 - **Function:** Dedicated chip that processes images, videos, and 3D graphics for display on the monitor.
- **Power Supply Unit (PSU)**
 - **Function:** Takes electricity from the wall outlet and converts it into power for all internal parts.
- **Cooling Fan / Heatsink**
 - **Function:** Draws heat away from sensitive components to prevent the computer from overheating.

Question 2: Steps to Disassemble a Desktop PC

Step 1 & 2: Unplugging Power & Opening the Side Panel



Step 1 & 2: Turned off the PC power supply, disconnected all main power cables, and unscrewed the side panel to open the cabinet.

Step 3 & 4: Removing RAM Modules



Caption: *Step 3 & 4: Unplugged the internal power supply wires and pushed open the plastic retention clips to remove the RAM stick from its slot.*

Step 5: Removing the Graphics Card (GPU)



Step 5: Unscrewed the rear mounting bracket, pressed the PCIe slot safety lock, and pulled out the graphics card.

Question 3: Air Cooling vs. Liquid Cooling

Factor	Air Cooling	Liquid Cooling
Cost	Cheap — Saves money and comes built-in with most PCs.	Expensive — Costs much more money to buy.
Cooling Power	Good — Fine for normal work, but can get hot and loud.	Excellent — Keeps the PC super cool and very quiet.
Maintenance	Easy — Just clean the dust off the fan sometimes.	Harder — Need to check for water leaks over time.

Question 4: Cooling Choice for Gaming & Streaming

- **My Choice: Liquid Cooling**

I choose **liquid cooling** because playing heavy games and streaming live at the same time makes the PC very hot. Liquid cooling keeps the system much cooler than regular fans so the PC runs fast without slowing down. It is also very quiet, so no loud fan noise will ruin my live stream audio.

3.

Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Question 1: (Reassembly)

Step 1: Put CPU, RAM, and SSD on the Main Board

- **What I did:** Put the processor (CPU), memory sticks (RAM), and storage (SSD) onto the motherboard while it was still outside the box.
- **Why this order:** It is much easier to snap small parts onto the board on a open table than inside a tight box.

Step 2: Attach the Fan and Put the Board Inside the Box

- **What I did:** Fixed the cooling fan on top of the CPU, placed the motherboard inside the PC box, and tightened the screws.
- **Why this order:** Fixing the fan first keeps the board safe, and mounting the board creates the base for everything else.

Step 3: Put in the Power Supply (PSU) and Hard Drive

- **What I did:** Screwed the power supply box into the bottom of the PC case and slid the hard drive into its slot.
- **Why this order:** Putting the power box in now lets you spread out the wires before big parts get in the way.

Step 4: Put in the Graphics Card (GPU)

- **What I did:** Pushed the graphics card into its slot until it clicked, then locked it with a screw at the back.
- **Why this order:** The graphics card is very big, so putting it in near the end stops it from blocking your hands.

Step 5: Plug In All Wires and Close the Box

- **What I did:** Connected all the power wires to the motherboard and graphics card, then slid the side cover back on and screwed it shut.
- **Why this order:** Wires are plugged in last so no loose cables hit the spinning fans or block air inside the box.

4.

Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

1. Checklist of Tools Needed

- **Phillips Head Screwdriver (+ Shape)**
 - **Why it is important:** Most PC screws (on the motherboard, graphics card, case, and power supply) use this exact size.
- **Anti-Static Wrist Strap**
 - **Why it is important:** It stops small electric shocks from your body from breaking delicate parts like the RAM or CPU.
- **Thermal Paste**
 - **Why it is important:** It fills tiny gaps between the processor and the cooler so heat moves quickly away from the CPU.
- **Zip Ties / Cable Ties**
 - **Why it is important:** They keep wires tidy so cables do not hit the spinning fans or block airflow.
- **Small Flashlight**
 - **Why it is important:** Helps you see dark corners and small pin labels inside the PC case clearly.

2. Checklist of Safety Rules

- **Unplug the Power Cable**
 - **Why it is important:** Prevents electric shocks and stops short circuits that can burn your PC parts.

- **Touch Metal Before Working (Grounding Yourself)**
 - **Why it is important:** Touching a metal object removes static electricity built up on your body so it won't damage computer chips.
- **Work on a Flat, Hard Table (Avoid Carpets)**
 - **Why it is important:** Working on wood or plastic stops static electricity from building up. Carpets create static easily.
- **Hold Parts by the Edges Only**
 - **Why it is important:** Never touch the shiny gold pins or open circuits on the parts because oil from your fingers can ruin them.
- **Keep Screws in Small Containers**
 - **Why it is important:** Prevents you from losing tiny screws or using the wrong screw in the wrong hole.